

Scalability & Future Plan

Date	2026-07-11
Team ID	
Project Name	Health History Organizer
Maximum Marks	1 Mark

Describe how the current solution can scale to support increasing users, hospitals, and healthcare records while outlining planned future enhancements.

Scalability

Aspect	Details
Current Architecture	FastAPI-based modular architecture with SQLAlchemy ORM, JWT authentication, RBAC, and SQLite/PostgreSQL database.
Scalability Approach	Containerization, database optimization, REST APIs, modular services, caching, and cloud deployment to support multiple hospitals.
Expected Growth / Load	Designed to support thousands of patients, multiple hospitals, concurrent users, and continuously growing health records.

Future Plan

S.No	Planned Feature / Enhancement	Target Timeline	Priority
1	Cloud-based deployment with automatic scaling	Phase II	High
2	HL7/FHIR integration for hospital interoperability	Phase II	High
3	AI-assisted disease prediction and clinical recommendations	Phase III	Medium
4	Android/iOS mobile application with wearable device integration	Phase III	Medium